

PROJECT PROPOSAL

SmartClinic: A Healthcare Appointment & Records Management System

CS 2005: Database Systems • Spring 2026

Course	CS 2005 – Database Systems
Submission	Project Proposal – 12th March, 2026
Project Title	SmartClinic: Healthcare Appointment & Records Management System

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1. Introduction and System Description

SmartClinic is a web-based Healthcare Appointment and Records Management System designed to digitize and streamline the day-to-day operations of a private clinic or small hospital. The healthcare sector is burdened by inefficiencies stemming from paper-based patient records, manual appointment scheduling, and fragmented communication between doctors, patients, and administrative staff.

SmartClinic addresses these challenges by providing a centralized, relational database-backed platform that manages patient registrations, doctor availability, appointment bookings, medical records, prescriptions, and billing — all through a clean and intuitive web interface. The goal is to reduce scheduling conflicts, eliminate data redundancy, and give every user role the right information at the right time.

2. System Functionality

SmartClinic will support the following core functions:

2.1 Patient Management

- Register new patients and maintain their profiles (demographics, contact info, blood type, allergies)
- View complete medical history and past visit records
- Search and filter patient records by name, ID, or date of visit

2.2 Appointment Scheduling

- Book, reschedule, and cancel appointments with available doctors
- View real-time doctor availability based on shift schedules
- Automated conflict detection to prevent double-booking

2.3 Medical Records & Prescriptions

- Doctors can create and update consultation notes after each visit
- Generate and store digital prescriptions linked to patient visits
- Track diagnosis history per patient over time

2.4 Billing & Invoicing

- Auto-generate invoices based on services rendered and prescriptions issued
- Track payment status (pending, paid, partially paid)
- Generate billing summaries and revenue reports for management

2.5 Staff & Doctor Management

- Manage doctor profiles, specializations, and shift schedules
- Administrative staff can manage system users and system-level settings

3. Users of the System

The system defines four distinct user roles, each with specific access rights and responsibilities:

User Role	Responsibilities & Permissions
Administrator	Manages all system users (create, update, deactivate accounts), oversees all data, generates high-level reports, and configures system settings. Has full access to all modules.
Doctor	Views assigned appointments, records consultation notes and diagnoses, issues prescriptions, and accesses patient medical history for their own patients.
Receptionist / Staff	Registers new patients, books and manages appointments, generates invoices, and handles patient check-in and check-out.
Patient	Logs in to view personal appointment history, medical records, prescriptions, and outstanding bills. Can also request or cancel appointments online.

4. Technologies Used

4.1 Front-End

- HTML & CSS – Page structure and styling
- Bootstrap – Responsive layout and pre-built UI components
- JavaScript – Client-side validation and dynamic UI interactions

4.2 Back-End

- Python(Flask) – Server-side logic, routing, and database connectivity

4.3 Database

- MySQL – Relational database for all persistent data storage
- SQL – Queries, views, stored procedures, triggers, and transactions

4.4 Development Tools

- Python venv – Virtual environment for managing project dependencies
- MySQL Workbench – Database design, ERD creation, and query testing
- VS Code – Source code editor

5. Scope of the System

5.1 In Scope

- Patient registration and profile management
- Appointment booking, rescheduling, and cancellation
- Doctor and staff account management
- Consultation notes and prescription management
- Billing and invoice generation
- Role-based access control (Admin, Doctor, Receptionist, Patient)
- Search and reporting functionality for each user role
- Data integrity enforcement via transactions and constraints

5.2 Out of Scope

- Integration with external healthcare systems or insurance providers
- Real-time video or telemedicine consultations
- Mobile application development (system is web-only)
- Advanced analytics or AI-driven diagnosis support
- Support for multi-branch or multi-hospital scenarios

6. Assumptions

- Each doctor operates within a single clinic and has a fixed weekly shift schedule.
- Each appointment is a one-to-one session between one doctor and one patient.
- Patients self-register online or are registered by a receptionist at the clinic.
- Payment is recorded manually by the receptionist; no payment gateway integration is included.
- All users access the system via a web browser on a local network or the internet.

7. Preliminary Entity Overview

The following entities are planned for the database design. Full normalization and ERD will be submitted in the final report.

- Patient – stores personal and medical background information
- Doctor – stores specialization, credentials, and shift availability
- Appointment – links patients and doctors with date, time, and status
- Medical Record – stores consultation notes and diagnoses per visit
- Prescription – stores medications prescribed during a visit
- Invoice – stores billing details per appointment
- User Account – stores login credentials and role assignments (links to Patient/Doctor/Staff)
- Staff / Receptionist – stores administrative staff profiles